

Problem 14

Prove that:

$$\bar{A}B + A\bar{B} + \bar{A}\bar{B} = \overline{AB}$$

Solution

We will prove this identity using Boolean algebra laws.

Method 1: Simplify the Left Side

Step 1: Factor Out Common Terms

$$\begin{aligned} & \bar{A}B + A\bar{B} + \bar{A}\bar{B} \\ \text{Group the first and third terms:} & \\ & = \bar{A}B + \bar{A}\bar{B} + A\bar{B} \\ & = \bar{A}(B + \bar{B}) + A\bar{B} \end{aligned}$$

Step 2: Use Complement Law

Since $B + \bar{B} = S$ (the certain event):

$$\begin{aligned} & = \bar{A} \cdot S + A\bar{B} \\ & = \bar{A} + A\bar{B} \end{aligned}$$

Step 3: Use Distributive/Absorption Property

We can rewrite this as:

$$\begin{aligned} \bar{A} + A\bar{B} &= \bar{A}(1) + A\bar{B} \\ &= \bar{A}(B + \bar{B}) + A\bar{B} \\ &= \bar{A}B + \bar{A}\bar{B} + A\bar{B} \end{aligned}$$

Wait, that's circular. Let me use a different approach.

Using the identity $X + \bar{X}Y = X + Y$:

$$\bar{A} + A\bar{B} = \bar{A} + \bar{B}$$

Step 4: Verify Using De Morgan's Law

By De Morgan's law:

$$\overline{AB} = \bar{A} + \bar{B}$$

Therefore:

$$\bar{A}B + A\bar{B} + \bar{A}\bar{B} = \bar{A} + \bar{B} = \overline{AB}$$

This completes the proof.

Method 2: Use De Morgan's Law on Right Side

Step 1: Expand the Right Side

$$\overline{AB} = \bar{A} + \bar{B} \quad (\text{De Morgan's law})$$

Step 2: Expand $\bar{A} + \bar{B}$

We need to show this equals $\bar{A}B + A\bar{B} + \bar{A}\bar{B}$.

$$\bar{A} + \bar{B} = \bar{A}(B + \bar{B}) + \bar{B}(A + \bar{A})$$

$$= \bar{A}B + \bar{A}\bar{B} + \bar{B}A + \bar{B}\bar{A}$$

$$= \bar{A}B + \bar{A}\bar{B} + A\bar{B} + \bar{A}\bar{B}$$

Using idempotent law ($X + X = X$):

$$= \bar{A}B + A\bar{B} + \bar{A}\bar{B}$$

This matches the left side.

Method 3: Truth Table Verification

Let's verify using a truth table:

A	B	$\bar{A}$	$\bar{B}$	$\bar{A}B$	$A\bar{B}$	$\bar{A}\bar{B}$	LHS
0	0	1	1	0	0	1	1
0	1	1	0	1	0	0	1
1	0	0	1	0	1	0	1
1	1	0	0	0	0	0	0

A	B	AB	$\overline{AB}$	RHS
0	0	0	1	1
0	1	0	1	1
1	0	0	1	1
1	1	1	0	0

The LHS and RHS columns match exactly.

Method 4: Algebraic Simplification (Detailed)**Step 1: Start with Left Side**

$$\bar{A}B + A\bar{B} + \bar{A}\bar{B}$$

Step 2: Factor $\bar{A}$ from First and Third Terms

$$= \bar{A}(B + \bar{B}) + A\bar{B}$$

Step 3: Apply $B + \bar{B} = S$

$$= \bar{A} \cdot S + A\bar{B} = \bar{A} + A\bar{B}$$

Step 4: Apply Absorption Identity

The identity states: $X + \bar{X}Y = X + Y$

With $X = \bar{A}$ and $Y = \bar{B}$:

$$\bar{A} + \bar{\bar{A}}\bar{B} = \bar{A} + \bar{B}$$

Since $\bar{\bar{A}} = A$:

$$\bar{A} + A\bar{B} = \bar{A} + \bar{B}$$

Step 5: Apply De Morgan's Law

$$\bar{A} + \bar{B} = \overline{AB}$$

Therefore:

$$\bar{A}B + A\bar{B} + \bar{A}\bar{B} = \overline{AB}$$

This completes the proof.

Interpretation

The left side represents: "(A doesn't occur and B occurs) OR (A occurs and B doesn't occur) OR (neither A nor B occurs)"

This is exactly the negation of " A and B both occur", which is the right side.

In other words, $\overline{AB}$ represents all scenarios except when both A and B occur simultaneously, which is precisely what the left side enumerates.

Venn Diagram Interpretation

- $\bar{A}B$: Region in B but outside A
- $A\bar{B}$: Region in A but outside B
- $\bar{A}\bar{B}$: Region outside both A and B
- Union of these three: Everything except $A \cap B$
- This is exactly $\overline{AB}$

Final Answer

$$\boxed{\bar{A}B + A\bar{B} + \bar{A}\bar{B} = \overline{AB}}$$

Proof (concise):

$$\begin{aligned}\bar{A}B + A\bar{B} + \bar{A}\bar{B} &= \bar{A}(B + \bar{B}) + A\bar{B} \\ &= \bar{A} + A\bar{B} \\ &= \bar{A} + \bar{B} \quad (\text{absorption}) \\ &= \overline{AB} \quad (\text{De Morgan})\end{aligned}$$

Key identities used:

- Complement law: $B + \bar{B} = S$
- Absorption: $X + \bar{X}Y = X + Y$
- De Morgan's law: $\overline{AB} = \bar{A} + \bar{B}$